

Learning an Agent-Centered Geometry of Quantum Possibility

Predictive State, Goal-Conditioned Control, and Directed Reachability from Measurement Experience Alone

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Abstract

We study a deliberately austere quantum-control problem. An artificial agent may choose a measurement and observe its classical outcome, but it is never given the quantum state, Kraus operators, Born probabilities, Hamiltonian, or tomographic estimates. Its experience is only a sequence of interventions and consequences. The scientific objective is therefore broader than quantum control: what operational representation of an unknown world can be learned from first-person experience, and can many possible aims be organized into a useful geometry?

We compare finite-history tabular Q-learning, goal-conditioned recurrent Q-learning, and a predictive multi-goal gated recurrent unit (GRU). The full agent jointly learns reward value, one-step outcome prediction, a stochastic-shortest-path cost, and goal embeddings regularized by policy similarity. Five environment families vary Hilbert-space dimension, hidden initial state, measurement strength, informational completeness, and outcome alphabet, including a qubit symmetric informationally complete (SIC) measurement and mutually unbiased qutrit bases. Thirty-six agents are trained across six scenarios, three backends, and two seeds.

The predictive geometry agent has the best macro-average success (80.5%), versus 79.1% for finite-history Q-learning and 77.7% for the reward-only GRU, but no backend dominates every environment and neural seed variance is large. Euclidean goal-embedding distance correlates moderately with held-out policy distance (mean Spearman $\rho = 0.54$) and more weakly with independent trajectory distance ($\rho = 0.31$). Learned reachability is directionally informative but not metrically calibrated: blank-history cost correlates $r = 0.47$ with empirical conditional hitting time and has mean absolute error 1.75 interventions. Intervention maps nevertheless reveal a legible directed structure in which outcomes reduce distance to matching goals while often increasing distance to alternatives. Goal geometry is thus a productive diagnostic and planning language, but its metric interpretation requires stronger validation.

Keywords: reinforcement learning; quantum measurement; partial observability; predictive state; goal-conditioned control; reachability; representation learning; QBism

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1 Introduction

Quantum control is usually described from the standpoint of an external modeler, who writes down a state ρ , a Hamiltonian, and control or measurement operators before optimizing a protocol. Model-free quantum control instead asks what can be learned through interaction, an approach already used for quantum feedback, state preparation, and error correction [18–20].

This work imposes a stricter informational boundary. The agent chooses an intervention a_t and receives a classical outcome o_t . It is not given ρ_t , measurement operators, expectation values, fidelities, or tomography. Its information at time t is

$$h_t = (a_1, o_1, \dots, a_{t-1}, o_{t-1}). \quad (1)$$

This interface is compatible with QBism’s emphasis on an agent’s actions, experiences, and expectations [1], although the algorithms do not assume or establish any interpretation of quantum theory. Agent-centered here describes the information boundary.

The central difficulty is partial observability. An external physicist may use ρ_t as a Markov state, but the learner cannot. It needs an internal summary $z_t = F(h_t)$ retaining distinctions relevant to prediction and control. This motivates three objects that must remain distinct:

1. **predictive state**: which histories imply the same future intervention/outcome statistics;
2. **strategic goal geometry**: which goals call for similar policies or trajectories;
3. **directed reachability**: how difficult a goal is from the agent’s present history.

Our methodological contribution is a validation framework comparing learned Euclidean goal embeddings with held-out policy behavior, trajectory signatures, empirical hitting times, finite-horizon success curves, and intervention-induced changes in all goal distances.

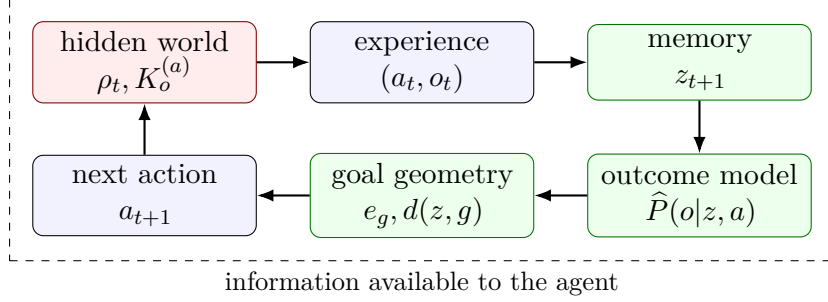


Figure 1: The operational learning loop. Quantum variables in red are private to the simulator. The agent constructs state, prediction, and goal-relative reachability from experienced action/outcome pairs.

2 Hidden quantum dynamics

2.1 Quantum instruments and backaction

A finite-dimensional quantum state is externally represented by a positive semidefinite unit-trace density operator ρ . A measurement action a has a Kraus operator $K_o^{(a)}$ for each classical outcome o , with physical completeness [2, 3]

$$\sum_o K_o^{(a)\dagger} K_o^{(a)} = I. \quad (2)$$

The hidden simulator samples and updates by

$$p(o|\rho, a) = \text{Tr}(K_o^{(a)} \rho K_o^{(a)\dagger}), \quad (3)$$

$$\rho' = \frac{K_o^{(a)} \rho K_o^{(a)\dagger}}{p(o|\rho, a)}. \quad (4)$$

Measurement therefore both supplies information and changes the conditions governing future experiences. This backaction is why reachability is naturally directed. The learner never evaluates equations (3) and (4); it receives only the sampled integer outcome.

For a projective basis $\{|b_o\rangle\}$, one may take $K_o = |b_o\rangle\langle b_o|$. The qubit worlds use Pauli Z , X , and Y bases. Their noncommutativity creates incompatible preparation strategies. The unsharp binary instruments use axis projectors P_+ , P_- and strength $q \in (1/2, 1)$:

$$K_0 = \sqrt{q} P_+ + \sqrt{1-q} P_-, \quad K_1 = \sqrt{1-q} P_+ + \sqrt{q} P_-. \quad (5)$$

At $q = 0.5$ the outcome is uninformative; as $q \rightarrow 1$ the operation approaches a projective measurement. We use $q = 0.8$.

A SIC-POVM in dimension d contains d^2 equiangular rank-one effects and is informationally complete [4]. For the qubit tetrahedral SIC,

$$E_i = \frac{I + \mathbf{r}_i \cdot \boldsymbol{\sigma}}{4}, \quad \sum_{i=0}^3 E_i = I, \quad (6)$$

where $\mathbf{r}_i$ are tetrahedral Bloch vectors; we use the positive square-root instrument $K_i = \sqrt{E_i}$. This tests mixed outcome alphabets: projective actions have two outcomes while the SIC has four. Invalid logits are masked.

Two bases $\{|e_j\rangle\}$ and $\{|f_k\rangle\}$ are mutually unbiased when $|\langle e_j | f_k \rangle|^2 = 1/d$ for all j, k . Certainty in one becomes a uniform distribution in the other. MUBs formalize complementarity and support tomography [5]. The qutrit world uses three projective MUBs.

Table 1: World/state scenarios. The hidden state is reset each episode.

Environment	d	Initial state	Measurement actions	Goals
qubit-zx-weak	2	$ 1\rangle$	projective Z, X ; weak Z	7
qubit-zx-weak	2	$ +\rangle$	projective Z, X ; weak Z	7
qubit-pauli	2	$ +i\rangle$	projective Z, X, Y	7
qubit-unsharp	2	$I/2$	weak Z, X, Y	6
qubit-pauli-sic	2	$I/2$	projective Z, X, Y ; tetrahedral SIC	9
qutrit-mub	3	$ 2\rangle$	three projective MUBs	8

2.2 Environment catalog

Operational state is relative to the available interventions. With an informationally incomplete repertoire, physically different density operators may imply identical statistics for everything an agent can do. Adding a SIC action or moving from qubit to qutrit may require a richer predictive representation. This motivates varying worlds rather than only random seeds.

3 Reinforcement learning from first principles

3.1 MDPs, return, and value

Reinforcement learning concerns sequential decisions with delayed consequences [6]. A finite Markov decision process is $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$: states, actions, a transition/reward law, and discount $\gamma \in [0, 1]$. The Markov property says the present state contains all past information needed to predict the next transition. A policy $\pi(a|s)$ is a distribution over actions. The discounted return is

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \quad (7)$$

The action value is $Q^\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$. Optimal value obeys the Bellman equation

$$Q^*(s, a) = \mathbb{E}[R_{t+1} + \gamma \max_{a'} Q^*(S_{t+1}, a') | s, a]. \quad (8)$$

The value of acting now is immediate reward plus the best discounted value available afterward.

3.2 Q-learning and exploration

Q-learning estimates [equation \(8\)](#) without knowing P or R [7]. From transition (s_t, a_t, r_t, s_{t+1}) , it forms target $y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a')$ (or r_t at termination) and updates

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[y_t - Q(s_t, a_t)]. \quad (9)$$

The bracket is the temporal-difference error. Q-learning is off-policy: the target assumes a greedy continuation even when behavior explores. With ϵ -greedy behavior, a uniformly random action is chosen with probability ϵ and a maximizing action otherwise. Classical convergence guarantees require finite tabular state/action spaces, suitable learning rates, and sufficient visitation [7]; they do not automatically extend to nonlinear recurrent networks.

3.3 Partial observability and predictive state

Here the simulator’s ρ_t could be Markov, but is hidden. The problem is a POMDP [8]; the policy is $\pi(a_t|h_t)$, not $\pi(a_t|\rho_t)$. The tabular baseline truncates experience to

$$s_t^{(L)} = (g, j_t, (a_{t-L}, o_{t-L}), \dots, (a_{t-1}, o_{t-1})), \quad (10)$$

where g is the goal and j_t its progress. This is transparent, but L is arbitrary, its state count grows exponentially, and similar histories share no strength unless identical.

Predictive-state representations instead characterize state by action-conditional predictions [9]. Ideally,

$$h \sim_{\text{pred}} h' \iff \mathbb{P}(o_{1:k}|a_{1:k}, h) = \mathbb{P}(o_{1:k}|a_{1:k}, h') \quad (11)$$

for every future test. Our recurrent vector is only an approximation, trained for control and one-step prediction. The predictive-state view supplies the interpretation and a future validation standard.

3.4 GRU memory and deep temporal-difference learning

A recurrent network maintains fixed-size memory

$$z_{t+1} = F_\theta(z_t, a_t, o_t). \quad (12)$$

With x_t the concatenated action/outcome embedding, a GRU [10] can be written

$$r_t = \sigma(W_r x_t + U_r z_t + b_r), \quad u_t = \sigma(W_u x_t + U_u z_t + b_u), \quad (13)$$

$$\tilde{z}_{t+1} = \tanh(W_h x_t + U_h(r_t \odot z_t) + b_h), \quad z_{t+1} = (1 - u_t) \odot \tilde{z}_{t+1} + u_t \odot z_t. \quad (14)$$

Reset and update gates learn what to forget and retain. Complete episodes are replayed through the GRU so late losses can alter earlier encoding: backpropagation through time. A slowly moving target network stabilizes TD targets, following deep Q-learning practice [11]:

$$\theta^- \leftarrow (1 - \tau)\theta^- + \tau\theta. \quad (15)$$

3.5 Evaluation principles

RL evaluation is difficult because training changes the data distribution and results vary across seeds. Reproducibility studies warn that few-run point estimates can reverse rankings [16, 17]. Evaluation should separate exploratory training from held-out greedy behavior, show per-task and aggregate results, and distinguish success probability from speed. Conditional duration can reward a reckless policy because failures are excluded. We therefore treat success as primary, label conditional steps, and retain full finite-horizon success curves. With two seeds, this study is exploratory rather than inferential.

4 Sequence goals and three geometries

4.1 Goal-conditioned control

A goal is an ordered sequence $g = ((a_1^g, o_1^g), \dots, (a_m^g, o_m^g))$. Other events may occur between checkpoints. Progress j_t is allowed because it is computed from the experienced history. Rewards are 10 for completion, 1 for an intermediate checkpoint, and -0.01 otherwise. A goal-conditioned value $Q(z, g, j, a)$ lets the same remembered history support different policies.

4.2 Symmetric strategic similarity

Each known goal has embedding $e_g \in \mathbb{R}^k$. Distance $D_E(g, h) = \|e_g - e_h\|_2$ is intended to mean similarity of strategy, not labels. At encountered state z , define

$$\pi_g(a|z, j) = \text{softmax}_a(Q(z, g, j, a)/T). \quad (16)$$

For policies p, q , Jensen–Shannon divergence is

$$D_{\text{JS}}(p, q) = \frac{1}{2}D_{\text{KL}}(p\|m) + \frac{1}{2}D_{\text{KL}}(q\|m), \quad m = \frac{1}{2}(p + q). \quad (17)$$

Its square root is a metric [13]. The regularizer is

$$\mathcal{L}_{\text{geom}} = \frac{1}{|\mathcal{P}|} \sum_{g < h} [\|e_g - e_h\|_2 - c\sqrt{D_{\text{JS}}(\pi_g, \pi_h)}]^2. \quad (18)$$

The policy target is detached. Geometry is therefore partly imposed. Held-out agreement tests generalization; it is not spontaneous emergence without bias.

4.3 Trajectory similarity

Local policy similarity may miss long-horizon differences. From held-out rollouts we build

$$\phi(g) = [\text{action frequencies, action/outcome frequencies, normalized duration, success rate}] \quad (19)$$

and $D_{\text{traj}}(g, h) = \|\phi(g) - \phi(h)\|_2$. This resembles successor features, which describe policies by expected future features and support transfer [14], although our signature is diagnostic rather than a learned successor model.

4.4 Directed reachability and displacement

Let τ_g be first completion time. The ideal intervention-count distance is

$$d^*(z, g, j) = \min_{\pi} \mathbb{E}_{\pi}[\tau_g|z, j]. \quad (20)$$

This is a stochastic shortest-path problem [12] with Bellman form

$$d^*(z, g, j) = 1 + \min_a \sum_o P(o|z, a) d^*(F(z, a, o), g, j'(a, o)). \quad (21)$$

The network learns positive action cost $C(z, g, j, a)$ and sets $d(z, g, j) = \min_a C(z, g, j, a)$. Its target is 1 on completion and otherwise

$$y_t^C = 1 + \min_{a'} C_{\theta^-}(z_{t+1}, g, j_{t+1}, a'). \quad (22)$$

This is directed: measurement can destroy conditions needed to return.

For N goals, the agent’s goal-relative position is

$$\mathbf{D}(z) = (d(z, g_1), \dots, d(z, g_N)). \quad (23)$$

An observed intervention produces displacement $\Delta \mathbf{D}_t = \mathbf{D}(z_{t+1}) - \mathbf{D}(z_t)$. Negative coordinates mean closer, positive farther. With the learned predictor, a contemplated action has

$$\mathbb{E}[d'_g|z, a] = \sum_o \hat{P}(o|z, a) d(F(z, a, o), g), \quad (24)$$

which provides a route to model-based planning.

A scalar mean hides risk, so we also estimate

$$R_g(z, T) = \mathbb{P}(\tau_g \leq T|z, \pi_g). \quad (25)$$

This parallels distributional RL’s emphasis on distributions rather than only expected return [15].

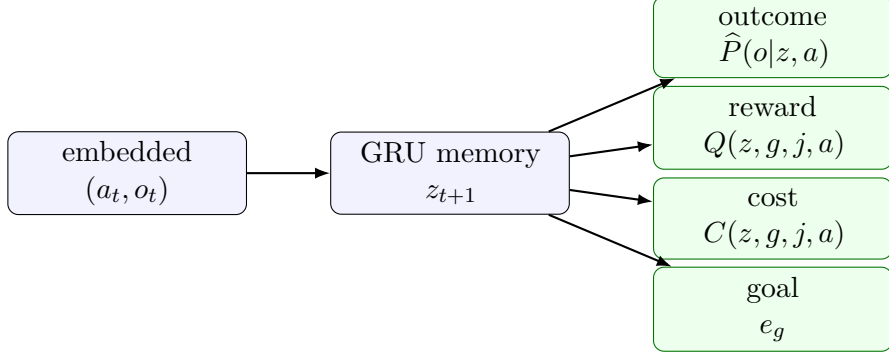


Figure 2: Shared recurrent representation and four learning objectives. Outcome prediction is goal-independent; reward and cost are goal-conditioned.

Table 2: Production configuration.

Training episodes/run	400	Evaluation episodes/goal	100
Geometry episodes/goal	40	Horizon	15
Discount γ	0.95	Tabular α	0.10
History length	4	Neural learning rate	10^{-3}
ϵ start/end/decay	1/.05/.99	Target τ	.02
GRU/goal dimensions	32/6	Interaction embedding	8
$\lambda_P, \lambda_C, \lambda_G$	.5, .5, .05	Geometry scale	2
Weak strength	.8	Seeds	0,1

5 Architecture and experimental protocol

All agents share one interaction protocol and receive only goal ID, progress, action, outcome, reward, and termination. The **tabular** state is [equation \(10\)](#) with $L = 4$. The **GRU** concatenates memory, a learned goal embedding, and one-hot progress before an MLP Q-head. The **predictive geometry GRU** adds a goal-independent outcome head, goal-conditioned cost head, and [equation \(18\)](#). Its prediction loss is

$$\mathcal{L}_{\text{pred}} = -\log \hat{P}_{\theta}(o_t|z_t, a_t), \quad (26)$$

computed before incorporating o_t . The full loss is

$$\mathcal{L} = \mathcal{L}_Q + \lambda_P \mathcal{L}_{\text{pred}} + \lambda_C \mathcal{L}_C + \lambda_G \mathcal{L}_{\text{geom}}. \quad (27)$$

Goals are sampled uniformly. Each episode resets hidden state, memory, and progress. Trajectories are replayed once for backpropagation through time; reward and cost use [equation \(15\)](#); gradients are clipped to norm 5.

Evaluation disables exploration. Separate geometry rollouts evaluate every goal-conditioned policy at each state, build trajectory signatures, compare blank-history cost to hitting time, estimate [equation \(25\)](#), and group displacements by action/outcome. PCA is used only to visualize embeddings; all distance comparisons use original dimensions.

The study contains 14,400 training episodes (107,804 interventions), 26,400 evaluation episodes (179,342 interventions), and 3,520 geometry episodes. An exact manifest, tidy CSVs, model weights, and plots are saved. Eighteen tests verify quantum validity, outcome masks, reproducibility, goals, runner semantics, every agent, and geometry utilities.

Table 3: Success rate, mean $\pm$ sample standard deviation over two seeds. Bold is the largest scenario mean.

World / state	Finite history	GRU	Predictive geometry GRU
Z/X/weak-Z / $ 1\rangle$	.810 $\pm$.012	.874 $\pm$.038	.861 $\pm$.035
Z/X/weak-Z / $ +\rangle$	.878 $\pm$.009	.809 $\pm$.063	.924 $\pm$.005
Pauli / $ +i\rangle$	.863 $\pm$.018	.806 $\pm$.057	.825 $\pm$.169
unsharp / $I/2$	.849 $\pm$.008	.788 $\pm$.080	.820 $\pm$.071
Pauli+SIC / $I/2$	.713 $\pm$.006	.649 $\pm$.027	.697 $\pm$.172
qutrit MUB / $ 2\rangle$	.634 $\pm$.034	.734 $\pm$.082	.703 $\pm$.015
Macro-average	.791	.777	.805

6 Results

6.1 Control performance

No backend dominates. Recurrent memory helps on the qutrit and original $|1\rangle$ scenarios; tabular learning leads on Pauli, unsharp, and Pauli+SIC; the full model is strongest from $|+\rangle$. Macro conditional steps are 4.62 (tabular), 4.55 (GRU), and 4.41 (full), but success remains primary.

Neural instability is substantial. Mean absolute seed gaps are 2.1 percentage points for tabular, 8.2 for GRU, and 11.0 for the full model; the largest full gap is 24.3. Both recurrent agents have zero success on the three-checkpoint qutrit goal, while tabular reaches 15%. The plain GRU also has zero success on the two-step $Z0_SIC0$ goal.

6.2 Goal-geometry validation

Across 12 full-agent runs, embedding distance has mean Spearman correlation 0.537 with held-out policy strategy distance (median 0.533, standard deviation 0.207, range 0.121–0.825). This is meaningful but imperfect generalization of the regularized relation. Correlation with more independent trajectory distance is weaker: mean 0.311, median 0.274, standard deviation 0.303, range -0.091 – 0.786 . Geometry is not yet a unique robust object. The first two PCA axes explain 77.3% of embedding variance on average.

6.3 Reachability and intervention displacement

Across 79 goals with at least one geometry-rollout success, predicted cost correlates $r = 0.471$ with conditional empirical steps and has mean absolute error 1.75. Predictions are compressed to 2.46–5.89 while observed means span 1.0–10.71. The head contains ordinal information but is not a calibrated clock. Off-policy minimum backups, function approximation, policy mismatch, finite-horizon censoring, and conditioning on success all contribute.

In [figure 8](#), $Z:0$, $X:0$, and $Y:0$ change matching-goal distance by -1.58 , -1.80 , and -1.91 . The four SIC outcomes change their matching distances by -2.43 to -2.95 , while most nonmatching coordinates increase. Completion explicitly sets a matching goal to zero, so diagonals should not be overinterpreted; structured off-diagonal change is the more interesting signature of backaction.

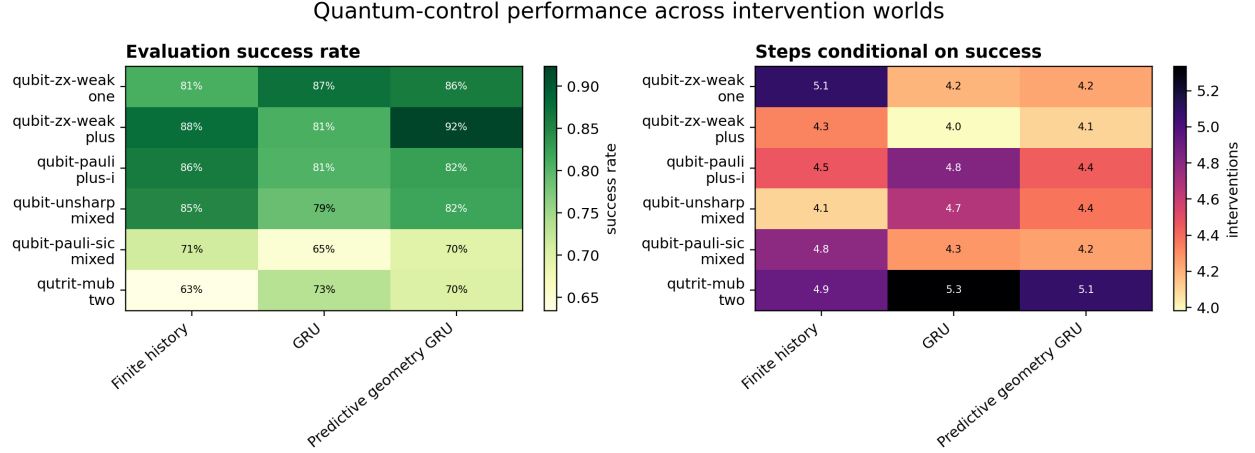


Figure 3: Held-out success and conditional intervention count, averaged over two seeds. Conditional steps exclude failures and must be read beside success.

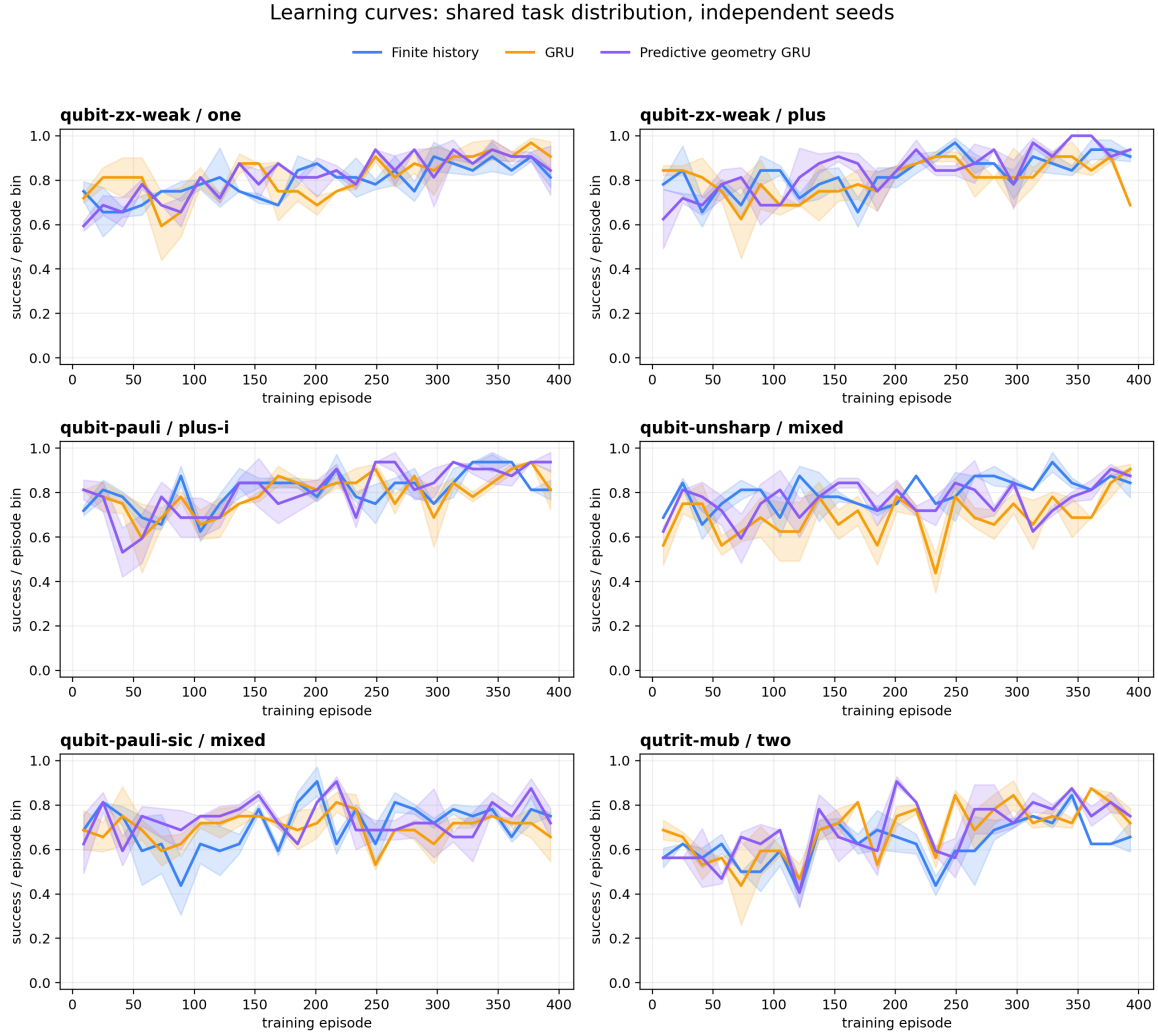


Figure 4: Binned exploratory training success. Shading is standard error over only two seeds and is descriptive, not a reliable interval.

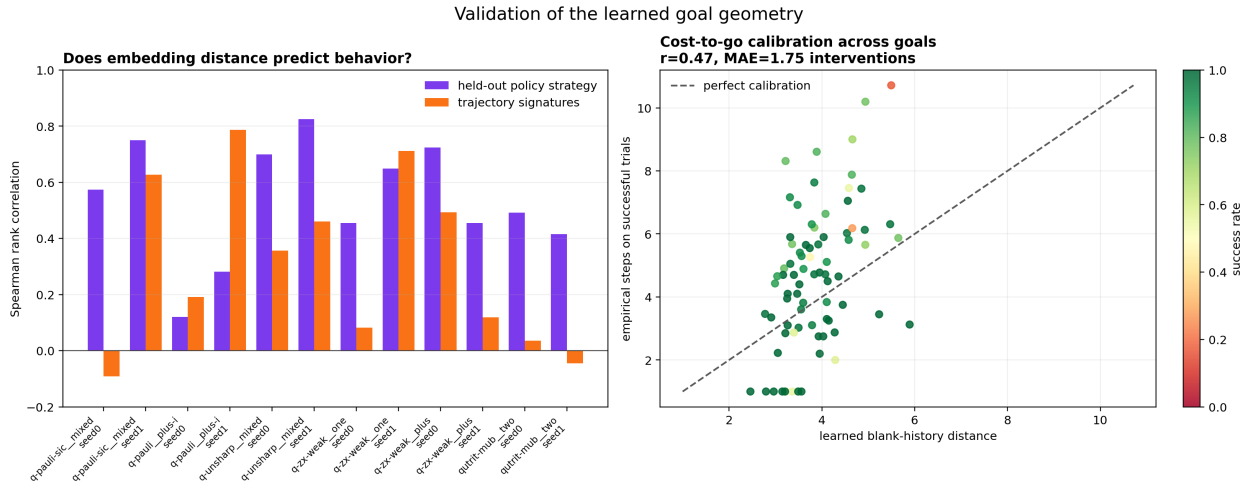


Figure 5: Left: correlation of embedding distance with held-out policy and trajectory distances. Right: cost-to-go calibration across successful goals.

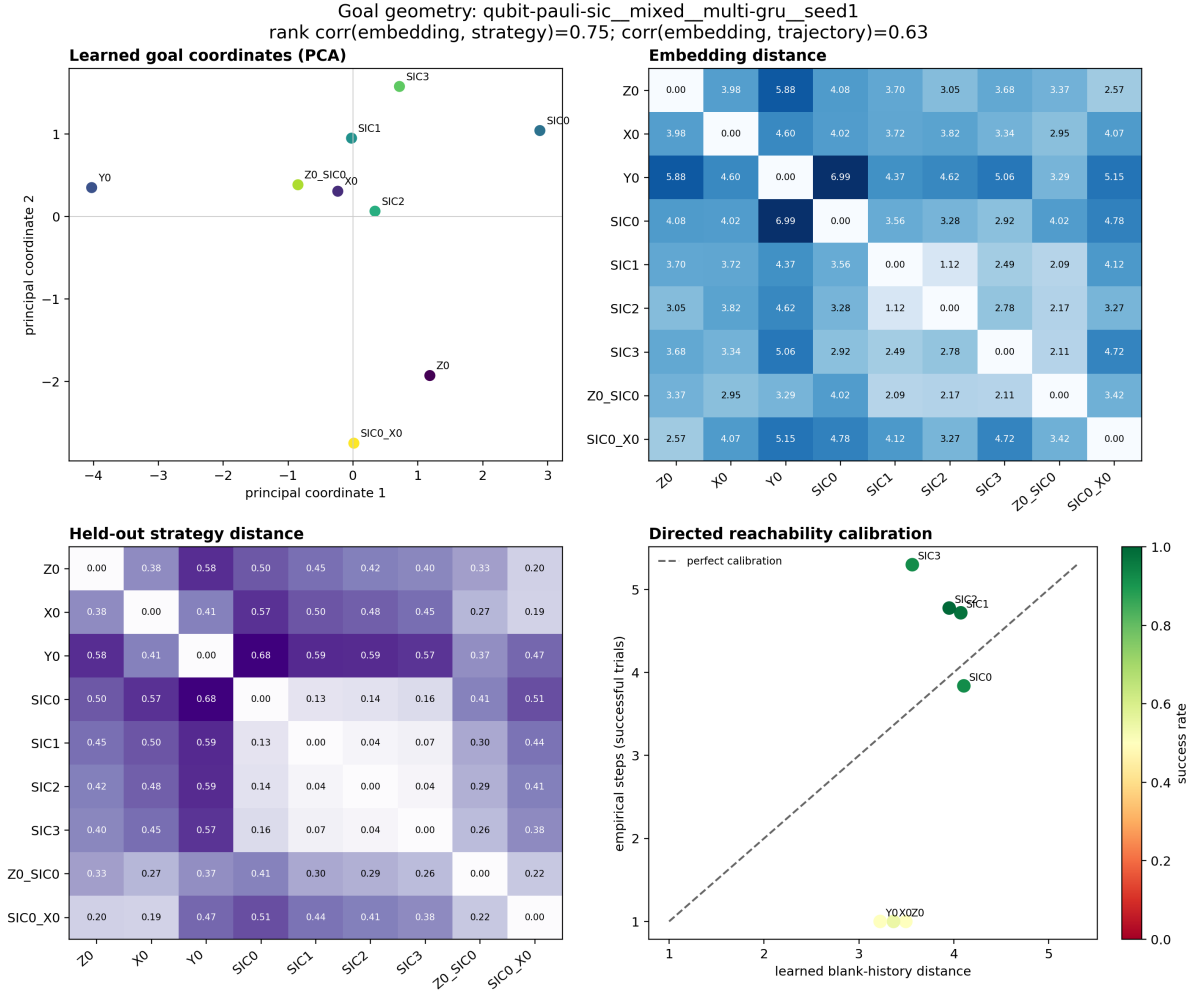


Figure 6: Representative Pauli+SIC geometry (seed 1): PCA coordinates, full embedding distances, held-out policy distances, and reachability calibration. The embedding has strategy correlation 0.75 and trajectory correlation 0.63 in this run.

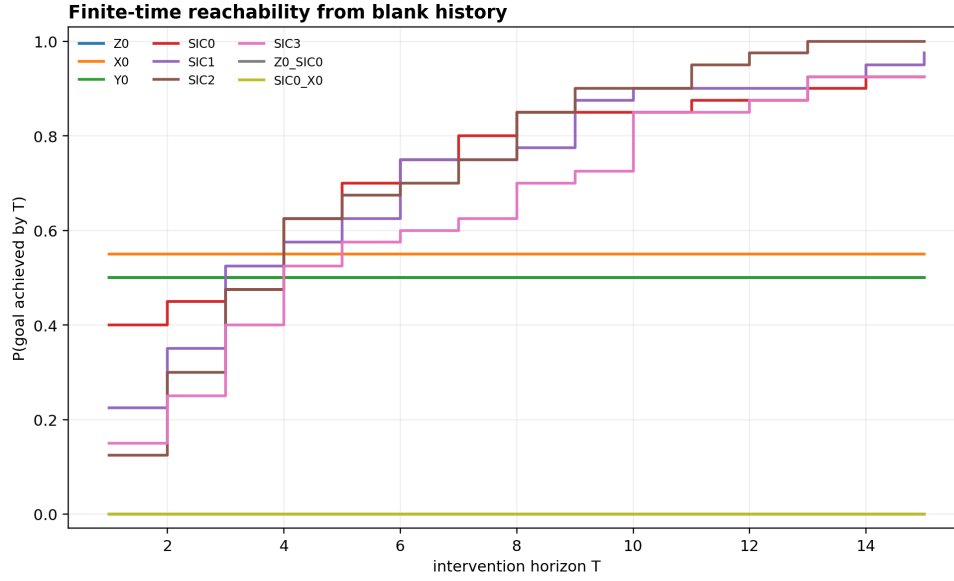


Figure 7: Finite-time success curves retain reliability information hidden by a conditional mean hitting time.

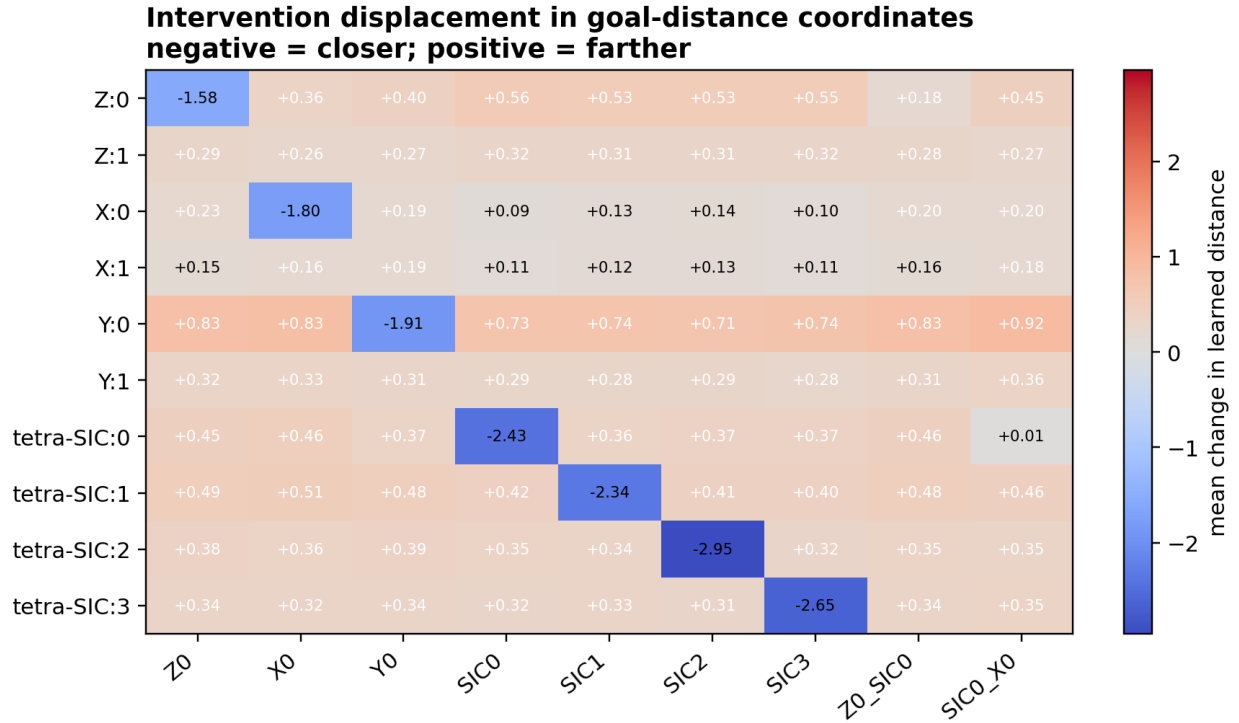


Figure 8: Mean displacement in learned goal-distance coordinates. Blue means closer and red farther. Matching outcomes create strong negative diagonals, while backaction alters multiple alternative affordances.

7 Discussion and limitations

Useful measurement strategies can be learned without exposing hidden quantum variables. Recurrent memory sometimes helps, but a short literal history is a strong small-world baseline. Goal embeddings carry moderate behavioral meaning, although local policy geometry transfers more reliably than whole-trajectory geometry. Directed coordinates expose coherent measurement displacements even though scalar cost is poorly calibrated.

These results do not show that a Bloch sphere, Hilbert space, or uniquely quantum geometry spontaneously emerged. Such a claim would require latent-state analysis, intervention ablations, classical controls, and privileged offline decoding of quantum invariants. Other limitations are:

- two seeds cannot support significance claims; modern RL methodology recommends more runs, intervals, robust aggregates, and performance profiles [17];
- episode budgets are equal, not compute budgets, and hyperparameters are not independently optimized;
- horizon 15 censors hard goals, reward shaping affects behavior, and conditional steps omit failures;
- one-step prediction does not guarantee predictive sufficiency;
- goal embeddings are lookup entries and cannot represent unseen goals;
- embedding targets come from the same Q-head, creating shared-error and circularity risks despite held-out evaluation;
- cost is learned off-policy from reward-oriented behavior and is not constrained to satisfy stochastic-shortest-path consistency;
- simulators omit Hamiltonian dynamics, decoherence, detector error, latency, and hardware drift.

8 Outlook

Compositional goals and transfer. Encode checkpoint sequences with a GRU or transformer, hold out complete goals, and test zero/few-shot transfer. Geometry matters if proximity predicts adaptation speed. Successor features provide a principled comparison [14].

Multi-step predictive state. Train action-conditional predictions over entire future tests, compare recurrent and transformer memory, and measure predictive calibration by horizon. Vary informational completeness to determine which latent dimensions disappear.

Planning in possibility space. Use [equation \(24\)](#) to compare direct Q-control, greedy distance reduction, and tree search through imagined outcomes. The learned geometry should improve decisions, not only visualization.

Distributional reachability. Learn τ_g distributions or $R_g(z, T)$ directly, using proper scoring rules and reliability plots. Distributional Bellman methods are a starting point [15].

Directed non-Euclidean structure. Explore directed graphs, asymmetric energy models, quasimetrics, hyperbolic goal hierarchies, and Finsler-like local costs. Choose geometry through predictive validity and planning utility, not aesthetics.

Information-seeking action. Replace undirected epsilon exploration with expected information gain, prediction uncertainty reduction, or reachability uncertainty reduction. This turns experimentation itself into goal-directed behavior.

Privileged offline physics. Pair learned z_t with hidden Bloch coordinates only after training. Test decodability, intrinsic dimension, convexity, and equivalence classes; compare against classical partially observed environments.

Stronger statistics and hardware. Use 10–20 seeds, longer budgets, bootstrap intervals, interquartile means, and performance profiles [16, 17]. Then add decoherence, readout error, continuous weak measurement, and device drift, ultimately learning from laboratory records as envisioned in model-free quantum control [19].

9 Conclusion

An agent beginning only with interventions and outcomes can learn more than a policy. It can compress history, predict contemplated measurements, compare aims by behavior, estimate goal-relative reachability, and represent measurement backaction as movement through future possibility. The present study makes that program concrete but also establishes a demanding standard: an attractive embedding is not self-validating. It must predict independent behavior, transfer, and reachability.

The geometry-aware agent is competitive but not universally superior. Its embedding has measurable, seed-dependent strategic meaning; its directed cost orders some goals but is not accurate in absolute units. These shortcomings identify the next experiments. The long-term prospect is an agent not handed a physical state space, but able to construct a predictive present and a geometry of possible aims from action and consequence alone.

A Reproduction and artifacts

```
conda run -n qbist_spacetime python run_experiment_suite.py \
  --scenarios qubit-zx-weak:one,qubit-zx-weak:plus,\
qubit-pauli:plus-i,qubit-unsharp:mixed,\
qubit-pauli-sic:mixed,qutrit-mub:two \
  --backends tabular,gru,multi-gru --seeds 0,1 \
  --episodes 400 --eval-episodes 100 --geometry-episodes 40 \
  --max-steps 15 --epsilon-decay 0.99 --device cpu \
  --output results/comparative
python plot_results.py results/comparative
```

The manifest records exact runs; episode CSVs retain raw outcomes; geometry directories store embeddings, three distance views, reachability, curves, and displacements; and the plot tree contains 51 figures.

B Goal repertoires

qubit-zx-weak

$Z0, Z1, X0, X1, Z0_X0, X0_Z0, weakZ0_Z0.$

qubit-pauli

$Z0, X0, Y0, Z0_X0, X0_Z0, Y0_X1, Z0_X0_Y0.$

qubit-unsharp

$wZ0, wX0, wY0, wZ0_wX0, wX0_wZ0, wZ0_wX0_wY0.$

qubit-pauli-sic

$Z0, X0, Y0, SIC0, SIC1, SIC2, SIC3, Z0_SIC0, SIC0_X0.$

qutrit-mub

$Z0, Z1, Z2, F0, F1, Z0_F0, F0_Z0, Z0_F0_P0.$

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